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15 NOVEMBER 1968
Superseding
MIL-I-7086A
21 December 1965

MILITARY SPECIFICATION

INDICATOR, PRESSURE, OIL, 0-100 PSI

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of hermetically sealed, remote indicating, MS28010-2, synchro-style oil pressure indicator.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-P-378	Plastic Sheet and Strip (Polyolefin)
PPP-C-843	Cushioning Material, Cellulosic

Military

MIL-P-116	Preservation, Methods of
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-T-7748	Transmitters, Pressure, Synchro, Aircraft, 320-Degree Movement, High Temperature, General Purpose
MIL-T-5796	Transmitter, Pressure, Oil, 0-100 PSI
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-20708	Synchros, General Specification for
MIL-T-25624	Transmitter, Pressure, Oil, Engine-Mounted, 0-100 PSI, Type TRU-80/A
MIL-B-27497	Bearing, Jewel, Sapphire, Synthetic

STANDARDS

Federal

Federal Standard No. 595	Colors

/FSC 6620/

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Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Requirements for Reliability Program (for Systems and Equipments)
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-834	Packaging Data Forms, Instructions for Preparation and Use of
MS28005	Transmitter, Pressure, Synchro, Aircraft
MS28010	Indicators, Pressure-Synchro, Single, 2-Inch Size
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard form of
MS33585	Pointer, Dial, Standard Design of Aircraft Instrument
MS33586	Metals, Definition of Dissimilar
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- The indicators furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.4 and 6.3).

3.2 Selection of specifications and standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.-

3.3.1 Nonmagnetic materials.- Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.2 Nonferrous materials.- Nonferrous materials shall be used for all parts of the indicator except where ferrous materials are essential.

3.3.3 Metals.- Metals shall be corrosion-resistant or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

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3.3.4 Dissimilar metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.3.5 Toxic and corrosive fumes.- The indicator shall be of such material and construction that, when burned, corrosive, deleterious or toxic fumes and gases that combine with the atmosphere to form acid or corrosive alkali will not be liberated.

3.3.6 Protective treatment.- When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.7 Bearing, synthetic.- When sapphire or ruby synthetic bearings are used, they shall be in accordance with MIL-B-27497.

3.4 Design and construction.- The indicator shall be designed for use with a TRU-80/A transmitter in accordance with MIL-T-25624, MIL-T-7748, or an MS28005 transmitter in accordance with MIL-T-5796.

3.4.1 Case.- The case shall be hermetically sealed and the dimensions shall conform to MS28010.

3.4.1.1 Cap.- A separate nonhermetically sealed cap may be used on the back of case for installing the electrical receptacle. The cap shall be made of nonferrous, density metal.

3.4.1.2 Case to cover glass distance.- The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 1/32 inch.

3.4.1.3 Filling medium.- The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain no more than 0.006 milligram of water vapor per liter (dewpoint -65° C) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi. Other concentrations of filling medium shall not be used except with prior approval of the procuring activity.

3.4.2 Pointer.- The pointer shall be in accordance with MS33585, except that the length of the pointer shall be such that the tip will extend into the scale a distance equal to one- to two-thirds the length of the shortest graduation.

3.4.2.1 Pointer finish.- The shaded portion of the pointer shall be finished in accordance with color No. 37875 of Federal Standard No. 595.

3.4.3 Dial.- The dial shall conform to figure 1. A minimum diameter of 1-3/4 inches shall be maintained across the outside ends of the graduations. The 30 pounds per square inch (psi) graduation shall be located on the horizontal centerline at the left side of the dial.

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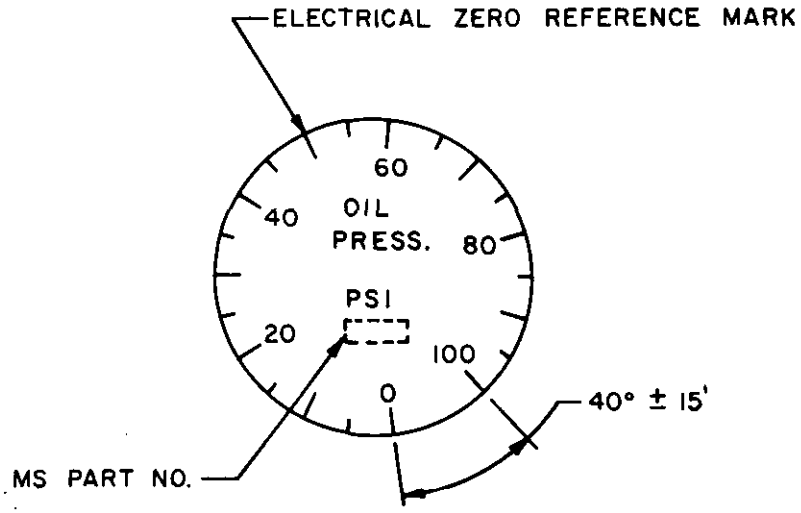


FIGURE 1. Dial

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3.4.3.1 Dial markings. - The markings specified in table I shall be finished in color No. 37875 of Federal Standard No. 595. The dimensions shall be as specified in table I.

TABLE I. Dial markings

	Height or length (inch ±./64)	Width of line or graduation (inch ±0.005)
Numerals 0, 20, 40, 60, 80, and 100	5/32	0.025
10-pound graduations	3/16	.031
Graduation at 50 psi	9/64	.031
Lettering OIL PRESSURE	1/8	.025
Lettering PSI	1/8	.025
5-pound graduations	3/32	.020

3.4.3.1.1 MS part number and dial background. - The applicable MS part number shall be permanently and legibly marked on the dial in letters 1/16-inch high. Location is optional. Unless otherwise specified, these markings and the background of the dial shall be finished in color No. 37038 of Federal Standard No. 595.

3.4.3.2 Dial visibility. - Portions of the minor graduations on the dial shall be visible when viewed from any position within the frustum of a cone the side of which intersects the outer edge of the cover glass and which makes an angle of 30 degrees with the perpendicular through the center of the dial.

3.4.3.3 Dial to cover glass distance. - The distance between the indicator dial and the inside of the cover glass shall be the minimum practicable and shall not exceed 0.125 inch.

3.4.4 Filler tube. - A tube or other facility for evacuating and filling the case shall be provided at the rear of the case and shall be recessed beneath the level of the back plate surface. If a tube is used, it shall be of such length as to permit three refillings of the case.

3.4.5 Cover glass. - The cover glass shall be clear, flat, and free from flaws that would interfere with the normal reading of the instrument. The thickness of the cover glass shall be the minimum practicable and entirely suitable for the purpose.

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3.4.6 Fasteners.- The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. If screws are used for attaching the dial, they shall be so located that they will not interfere with dial markings or prevent the markings from being located in the proper position.

3.4.7 Electrical zero reference mark.- The dial shall be provided with an electrical zero reference mark on the scale portion equidistant in angular degrees between the first and last dial graduations.

3.4.8 Electrical connector.- The indicator shall be provided with an electrical connector conforming to MS33678-14S-2P, or equal, and shall be hermetically sealed. The electrical connector shall have the polarizing key at the top.

3.4.9 Electrical wiring.- The external wiring of the indicator shall be in accordance with figure 2.

3.4.10 Indicator synchro.- The indicator synchro shall consist of a salient two-pole wound rotor used as the primary and shall be designed for 26V, 400 Hz, as required, single-phase power, with wye-connected stators as the secondaries. The stators shall be so wound that the amplitudes of the induced voltages measured between any two stator leads will vary approximately sinusoidally with the angular positions of the rotor. Pivots, bearings, and gears shall not bind nor shake and shall be as nearly frictionless as possible.

3.4.10.1 The synchro receiver used shall have electrical characteristics compatible with those defined in MIL-S-20708.

3.4.10.2 Secondary voltage.- The maximum open-circuit voltage measured between any two secondary leads shall be 11.8V ±3 percent.

3.4.11 Voltage and frequency variations.- The indicator shall withstand voltage and frequency variations from 22V to 30V and 380 Hz to 420 Hz, without exceeding the scale error and friction tolerances (see 3.5.1 and 3.5.2).

3.4.12 Electromagnetic interference.- The indicator shall meet the radiated and conducted electromagnetic interference requirements of MIL-STD-461 for class ID equipment.

3.4.13 Reliability program.- The contractor shall establish a reliability program in accordance with MIL-STD-785.

3.4.13.1 Reliability.- The indicator shall have a minimum acceptable mean-time-between-failures (MTBF) of 2,000 hours based upon a discrimination ratio of 1.5 and a confidence factor of 90 percent.

3.4.14 Longevity.- The indicator shall have an operating life span of 3,000 hours.

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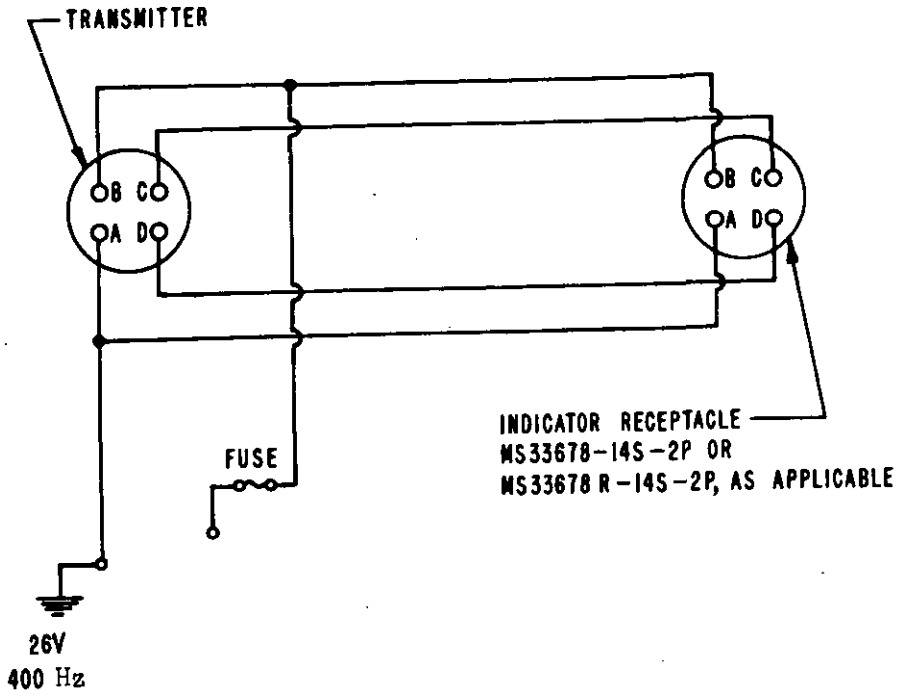


FIGURE 2. Wiring diagram

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3.5 Performance. - The indicator shall operate satisfactorily after subjection to the following conditions:

(a) Temperatures - temperatures ranging from -65° to +71° C
(b) Vibration - vibration of 2g's from 5 to 500 Hz
(c) Temperature-altitude-pressure altitude of 100,000 feet at -55° C
(d) Thermal shock - eight cycles of alternate immersions in tap water at 85° ±2° C and 5° ±2° C
(e) Salt fog - exposure to salt sea atmosphere for 48 hours without evidence of corrosion or deterioration

3.5.1 Scale error. - Scale error shall not exceed the tolerances specified in tables II and III.

TABLE II. Tolerances - scale error

PSI	Tolerance ± psi
0	1.0
20	1.0
40	1.0
60	1.0
80	1.0
100	1.0

TABLE III. Test tolerances

Test	Tolerance ± psi
Position error at midscale	1.0
Friction	
Room temperature	1.0
High temperature	1.0
Low temperature	1.0
Vibration	
Pointer oscillation	1.0
Pointer variation	1.0
Scale error	
Low temperature	1.0
High temperature	

3.5.2 Friction error. - Friction error shall not exceed the tolerances specified in table III.

3.5.3 Position error. - Position error shall not exceed the tolerance specified in table III.

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3.5.4 Dielectric strength. - The indicator shall withstand 200V rms at a commercial frequency for a period of 5 seconds.

3.5.5 Case leakage. - Case leakage shall not exceed 0.1 micron cubic foot per hour.

3.5.6 Fogging. - There shall be no evidence of moisture or oil fogging of the cover glass when the indicator is subjected to a temperature of 85° ±5° C for a minimum period of 2 hours and then the face of the cover glass immediately immersed in an ice bath for 3 minutes.

3.5.7 Effect of lead length. - When the lead length between the indicator and transmitter is changed from 5 to 500 feet, the indication shall not change more than 1 degree of arc. The ground lead shall be kept short.

3.5.8 Magnetic effect. - Compass deflection shall not exceed 3 degrees of arc when the indicator is positioned 5-1/2 inches from the compass and the horizontal magnetic field intensity at the indicator is 0.17 to 0.19 oersted.

3.5.9 Unit torque gradient. - The unit torque gradient shall be a minimum of 2,000 milligram-millimeters per angular degree of rotation.

3.6 Part numbering of interchangeable parts. - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Numerals and letters. - Numerals and letters shall conform to MS33558.

3.8 Finishes and protective coatings. -

3.8.1 Aluminum alloy parts. - Aluminum alloy parts shall be coated with an anodic film conforming to MIL-A-8625 except that small holes, straight threads, and case inserts need not be anodized.

3.9 Weight. - The weight of the indicator shall not exceed 0.60 pound.

3.10 Screw threads. - Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.11 Identification of product. - Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.11.1 Nameplate. - A nameplate shall be permanently attached to the rear of the case. Where applicable, the information marked thereon shall conform to MIL-STD-130. The nameplate shall not be glued or cemented to the case nor shall metal decals be used.

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3.12 Workmanship. - The indicator shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges. Gears shall be of high quality and shall provide a minimum of backlash necessary for smooth operation. Where necessary, antibacklash techniques shall be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification if inspections.- The examination and testing of indicators shall be classified as follows:

(a) Qualification inspection (4.4)
(b) Quality conformance inspection (4.5)

4.3 Test conditions. -

4.3.1 Standard atmospheric conditions. - Wherever the pressure and temperature exisitng at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg.) and at room temperature (approximately 25° C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Attitude. - Unless otherwise specified, the indicator shall be tested in the normal operating position with the plane of the dial vertical.

4.3.3 Electrical power. - Unless otherwise specified, the indicator shall be tested with 26 ±1V, single-phase, 400 ±5 Hz ac power.

4.3.4 Voltage and frequency variation. - When performing the scale and friction error tests specified in 4.6.4, the voltage and frequency variation shall be from 22V to 30V and 380 Hz to 420 Hz, or any combination thereof.

4.3.5 Tapping. - Unless otherwise specified, the indicator shall be lightly tapped or vibrated before each reading.

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4.3.6 Test tolerances. - The indicator shall be operated within the tolerances specified in tables II and III.

4.4 Qualification inspection. - The qualification inspection shall consist of the examination and tests specified in 4.6.

4.4.1 Test samples. - The test samples shall consist of 10 indicators representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required (see 6.2 and 6.3). Seven of the samples shall be subjected to the individual tests (see 4.5.1) and the reliability test (see 4.6.21), and the other three shall be subjected to all the tests except reliability.

4.4.2 Test report. - The contractor shall prepare and submit a test report in accordance with MIL-STD-831.

4.4.3 Nonstandard parts data approval. - When nonstandard electronic parts are used in the construction of the indicator, nonstandard parts data approval shall be a prerequisite to qualification approval.

4.5 Quality conformance inspection. - Quality conformance inspection shall consist of the following:

(a) Individual tests (4.5.1)
(b) Sampling tests (4.5.2)

4.5.1 Individual tests. - Each indicator shall be subjected to the following examination and tests:

(a)	Examination of product	(4.6.1)
(b)	Dielectric strength	(4.6.2)
(c)	Electrical zero	(4.6.3)
(d)	Scale and friction error	(4.6.4)
(e)	Position error	(4.6.5)
(f)	Sealing	(4.6.6)

4.5.2 Sampling tests. -

4.5.2.1 Sampling plan A. - One indicator shall be selected at random, from each 100 or less produced on the contract or order, and subjected to the following tests:

(a)	Individual tests	(4.5.1)
(b)	Vibration	(4.6.7)
(c)	Low temperature	(4.6.8)
(d)	High temperature	(4.6.9)
(e)	Fogging	(4.6.10)

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4.5.2.2 Sampling plan B. - Unless otherwise specified (see 6.2), three indicators shall be selected at random from the first 15 on the contract or order and subjected to the following tests:

(a)	Sampling plan A tests	(4.5.2.1)
(b)	Secondary voltage	(4.6.11)
(c)	Magnetic effect	(4.6.12)
(d)	Effect of lead length	(4.6.13)
(e)	Temperature-altitude	(4.6.14)
(f)	Low temperature exposure	(4.6.15)
(g)	High temperature exposure	(4.6.16)
(h)	Thermal shock	(4.6.17)
(i)	Salt fog	(4.6.18)
(j)	Electromagnetic interference	(4.6.19)
(k)	Unit torque gradient	(4.6.20)

4.5.2.3 Rejection and retest. - When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of the failure have been determined and appropriately corrected.

4.5.2.4 Individual tests may continue. - For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. - The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods. -

4.6.1 Examination of product. - The indicator shall be inspected to determine conpliance with this specification with respect to materials, workmanship, and dimensions. Prior to assembly, dials, pointers, and synchros shall be inspected as to markings, dimensions, and finishes.

4.6.2 Dielectric strength. - A potential of 200V at a commercial frequency shall be applied between any pin of the electrical receptacle and the case of the indicator for a minimum period of 5 seconds. There shall be no breakdown of the insulation.

4.6.3 Electrical zero. - The electrical zero test and the verification of leads shall be accomplished on the completed indicator with the case removed. A voltage of 26 ±1V single phase shall be applied to the rotor of the synchro with leads designated R_1 and R_2. R_2 shall be connected to the ground or low potential side of the power input. With a lead designated S_2 connected to R_2, the following conditions shall exist at electrical zero:

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(a) The voltage between the remaining stator terminals designated S_1 and S_3 shall be a minimum.
(b) The voltage between R_1 and S_3 shall be near the low null and shall reach the low null when the rotor is rotated clockwise approximately 30 degrees from electrical zero.

4.6.4 Scale and friction error. - The indicator shall be tested for scale error at ambient temperature at the test points specified in table III. The indicator shall be lightly tapped or vibrated before each reading to remove normal instrument friction. The scale errors shall not exceed the tolerances specified in table II. The friction error test may be conducted in conjunction with the scale error test. The difference between the readings before and after the indicator is tapped or vibrated shall constitute friction error. The friction error shall not exceed the tolerances specified in table III. This procedure shall be accomplished both up-scale and down-scale at the test points specified.

4.6.5 Position error. - The indicator pointer shall be positioned at the electrical zero reference mark with power applied. The change in pointer position when the indicator is held in any position shall not exceed the value specified in table III.

4.6.6 Sealing. - The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere.

4.6.7 Vibration. - The indicator shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514, equipment category (a), Procedure I, curve B. A synchro signal shall be maintained on the stators of the indicator synchro to provide an indication of 45 psi. The pointer oscillation shall not exceed ± 1 psi and pointer variation shall not exceed ± 1 psi (see table III). Upon completion of this test, the indicator shall be subjected to the scale error test at room temperature. The scale errors shall not exceed the tolerances specified in table II.

4.6.8 Low temperature. - The indicator shall be subjected to a temperature of -55° ± 2° C for a period of 4 hours. While at this temperature, it shall be checked for friction and scale errors and shall not exceed those specified in table III. In every case, the indicator shall not require more than 3 seconds to come within the tolerance specified.

4.6.9 High temperature. - The indicator shall be subjected to a temperature of 71° C for a period of 4 hours. While at this temperature, it shall be checked for friction and scale errors. The scale and friction error tolerances shall not exceed the values specified in table III. In every case, the indicator shall not require more than 3 seconds to come within the tolerance specified.

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4.6.10 Fogging. - The indicator shall be placed in a controlled ambient temperatuer of 85° ±5° C for a minimum period of 2 hours. Within 10 seconds after removal from the chamber, the indicator shall be immersed to a depth of 1/4 inch in an ice water bath maintained at 0° to 7° C. After 3 minutes, the indicator shall be removed from the fluid bath and quickly wiped dry (compressed air shall not be used). The cover glass and dial shall immediately be inspected and any evidence of fogging or damage shall be cause for rejection.

4.6.11 Secondary voltage. - The maximum open-circuit voltage shall be measured between any two secondary leads and shall be 11.8V ±3 percent. The variation between the three maximum secondary voltages on the indicator synchro as measured between any two stator leads shall not exceed 0.2V.

4.6.12 Magnetic effect. - The indicator, first, not operating and second, operating on rated voltage, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. Deflection of the compass at any of the specified positions shall not exceed 3 degrees.

4.6.13 Effect of lead length. - The indicator shall not change more than 1 degree of arc when the lead length between the indicator and transmitter is changed from 5 to 500 feet of No. 20 AWG gage wire. Equivalent resistances may be substituted for the added wire length. The ground lead shall be kept short.

4.6.14 Temperature-altitude. - The temperature-altitude test shall be performed in accordance with MIL-STD-810, Method 504, Procedure I (except the temperature shall be -55° C with a chamber pressure equivalent to an altitude of 100,000 feet. The rate of pressure change shall not exceed an average of 3 psi per second while under these conditions. At the end of a 1-hour period, the scale and friction error test shall be performed. The errors shall not exceed the tolerances specified in table III. This test may be performed concurrently with the low temperature exposure test (see 4.6.15).

4.6.15 Low temperature exposure. - The indicator shall be subjected to a temperature of -65° ±2° C for a period of 48 hours. While at this temperature, the indicator shall be operated to assure that the pointer moves over the entire scale range without exhibiting sticky or erratic characteristics. The temperature shall then be raised to -55° ±2° C and held for a period of 24 hours. While at this temperature, the indicator shall be subjected to the scale error and friction test, except that in every case, the indicator shall not require more than 4 seconds to come within the tolerances of table III. All parts of the indicator shall then be maintained at room temperature for a period of not less than 4 hours and the indicator again subjected to the scale error and friction test. The scale errors shall not exceed the tolerances specified in table II. There shall be no damage or deterioration that will affect subsequent operation.

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4.6.16 High temperature exposure.- The indicator shall be subjected to a temperature of 70° ±2° C for a period of 24 hours and then tested at this temperature for scale errors as specified in 4.6.4. The scale error and friction shall not exceed the tolerances specified in table III. All parts of the indicator shall then be maintained at room temperature for a period of not less than 4 hours and the indicator again subjected to the scale error and friction test. The scale errors observed shall not exceed the tolerances specified in table II. There shall be no damage or deterioration that might affect subsequent operation.

4.6.17 Thermal shock.- The indicator shall be subjected to eight cycles of immersion in water at a temperature of 85° ±2° C and then in water at a temperature of 5° ±2° C without evidence of moisture penetration or damage to the coating or the case. The length of time for each immersion shall be 30 minutes, with not more than 5 seconds between immersions. After eight continuous cycles have been completed, the indicator shall be subjected to the test specified in 4.6.6.

4.6.18 Salt fog.- The indicator shall be subjected to a salt fog test for a period of 48 hours in accordance with MIL-STD-810, Method 509, Procedure I. There shall be no corrosion or deterioration that might affect subsequent operation. After exposure to salt fog, the indicator shall be subjected to the tests specified in 4.6.4 and 4.6.6.

4.6.19 Electromagnetic interference.- The indicator shall be subjected to an electromagnetic interference test in accordance with MIL-STD-462. The test methods shall be as specified in MIL-STD-461 for class ID equipment.

4.6.20 Unit torque gradient.- The indicator shall be connected to the [illegible]ter transmitter synchro and the master transmitter locked in the electrical zero position. The synchro rotors shall then be energized with rated voltage and frequency ±1 percent. Weights shall be applied to a torque pulley which is positioned on the rotor shaft to rotate the shaft in a counterclockwise direction through an angle of 5 degrees of use. The torque tending to return the shaft to its original position shall be computed and shall be no less than 2,000 milligram-millimeters per degree of displacement.

4.6.21 Reliability.- Seven indicators shall be tested for reliability. Acceptance permitting production and total test time shall be in accordance with MIL-STD-781, test plan I. Recording, data handling, and reporting shall also be in accordance with MIL-STD-781.

4.6.21.1 Test.- The test equipment and fixtures shall include provisions for vibration and temperature cycling at the periodic intervals specified herein. During this test, input signals shall be applied to the indicator at a rate of 2 to 3 cycles (minimum and maximum) per minute from 0 to 85 percent of full scale and return. The test shall begin when rated power is applied. After completion of 4 hours of operation with rated power on, which shall include at least 480 cycles of operation, the power shall be turned off and shall remain off for a period of 10 minutes. For at least 10 minutes out of every hour of ON time, the indicator shall be vibrated in accordance with test level J of MIL-STD-781. The temperature cycles shall be performed in the following sequence: 4 hours at -20° C, 14 minutes for temperature change to +50° C at a minimum rate of 5° C per minute; 4 hours at +50° C, 14 minutes for temperature change to -20° C, and 4 hours at -20° C. In case of malfunction, the malfunction shall be recorded and the test

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interrupted while repairs are made. Upon completion of each 250 hours of cycling, the indicator shall be checked for scale error. The scale error reading shall not exceed 150 percent of the room temperature tolerance in table III. No recalibration shall be performed on the units during this test. The test may be interrupted for weekend shutdown. The contractor shall prepare a report in accordance with MIL-STD-781 which shall contain test procedures, details of any malfunctions, and corrective actions taken.

4.6.22 Longevity.- At the completion of the reliability test, testing of three of the indicators shall be continued for a total of 3,000 hours each. The test time accumulated during the reliability test shall be applied to this test. The accept-reject criteria is not applicable.

4.7 Inspection and test.- Tests of methods of preservation and packaging shall be accomplished in accordance with section 4 of MIL-P-116 to insure compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Each indicator shall be preserved and packaged in quantity unit packs of one each in accordance with Method 1C-2 of MIL-P-116. The indicator shall be wrapped with polyfilm L-P-378, or equal, and cushioned in accordance with PPP-C-843 prior to placing in the container required by the method of preservation. The above requirements apply for stock items, unless otherwise specified in the contract or order (see 6.2).

5.2 Packing.- Each indicator procured in accordance with this specification shall be packaged level B in accordance with MIL-STD-794, unless otherwise specified in the contract or order (see 6.2).

5.3 Marking.- In addition to any other marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

INDICATOR, PRESSURE, OIL, 0-100 PSI, MS28010-2

5.3.1 Reinspection date.- Reinspection date markings shall be in accordance with the instructions of the procuring activity (see 6.2).

6. NOTES

6.1 Intended use.- The indicator covered by this specification is intended for use in conjunction with the TRU-80/A or an MS28005 oil pressure transmitter in indicating remotely the oil pressure in aircraft systems which require integrally lighted instruments.

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6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Information, if any, to be submitted with qualification samples (4.4.1)
(c) When sampling plan B tests will not be conducted (4.5.2.2).
(d) Levels of packaging and packing required, and reinspection date markings (see section 5).

6.3 Qualification. - With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Aeronautical Systems Division (ASD), Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

6.4 International standardization agreement. - Certain provisions (3.3.3 and figure 1) of this specification are the subject of international standardization agreement ASCC 10/2 B. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.5 Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Air Force - 11
Army - EL
Navy - AS

Reviewer activities:
Air Force - 11, 71
Army - MC, MO
Navy - AS

International interest (see 6.4)

Preparing activity:
Air Force - 11

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